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Computer Science (BS)


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Affiliation: School of Science and Mathematics

The goal of the bachelor's degree program in computer science is to educate a graduate with a broad intellectual base, well-developed interpersonal, analytical and problem-solving skills, and a mastery of the appropriate elements of the discipline of computer science. Graduates should be well-qualified for advanced study in strong graduate programs or employment in a professional career requiring extensive preparation in computer science.

THE DEGREE PROGRAM

The major requirements of the BS degree program for computer science majors are consistent with the curriculum recommendations of the Association for Computing Machinery, the Computer Society of IEEE, and ABET accreditation standards. In addition to the University's *Dialogues*, the major requires support including two semesters of calculus, one semester of statistics, and an additional STEM course.

Consistent with the ABET criteria, Computer Science Graduates will have an ability to:

1. Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.
2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
3. Communicate effectively in a variety of professional contexts.
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.

Computer Science graduates will be well-qualified for advanced study in graduate programs or employment in a professional computing career. All CS graduates will complete a sequence of courses to learn coding and computing fundamentals, courses that examine relevant mathematical structures, theoretical foundations of computing, systems programming, computer architecture, data structures, applications and project management, and the ethics of computing, as well as electives that allow them to customize their program.

A capstone experience gives students the opportunity to integrate the areas of knowledge acquired in the major on a research project, internship, or other substantial project in software development or a related area. Students also complete two seminars that focus on understanding the broader field, career

and graduate school opportunities, and to develop practical skills and attitudes that will help students find success after graduation.

COMPUTER SCIENCE COMMUNICATION

All computer science majors will develop their skills as speakers and as writers through multiple communication experiences.

- **CS 345 - Cyberethics** is a Writing-Enhanced course where students examine “writing about research” and are required to give presentations.
- **CS 370 - Software Engineering**, which relies on a project-based approach to explore deliverables including written work and presentations.
- **CS 497 - Computer Science Capstone Experience** (or similar) also includes a presentation and a paper.

Other elective CS courses involve writing and speaking experiences in a variety of settings, including longer and shorter forms, formal and informal styles, in-person and online delivery, with a special emphasis on communication products aimed at technical experts and non-experts.

Overview of the Computer Science Introductory Courses

Computer science majors take **CS 180 - Foundations of Computer Science I** and **CS 181 - Foundations of Computer Science II**, **CS 191 - Computing Structures**, and **CS 291 - Models of Computation**. This collection of courses is an introduction to computer science and to the mathematical foundations of computing.

A non-major introductory course, **CS 170 - Introduction to Computer Science I**, is offered for those who want an introduction to the field, but do not plan to take additional CS courses. Non-majors are also welcome in the CS 180 course. All those who want to major or minor in Computer Science should take CS 180. A student who takes CS 170 and then decides to continue with CS courses is required to take CS 180. For Computer Science minors, CS 170 can be counted toward the minor requirement.

ENRICHMENT OPPORTUNITIES

Several opportunities are available for students to pursue their interests and become involved in learning communities, and to get acquainted with peers and faculty outside classroom settings. A student chapter of the Association for Computing Machinery, a national professional society for computer scientists, hosts guest speakers, capstone presentations and works on service projects. The Tru Women in Computer Science (TWiCS) is an organization of women students who have an interest in computer science. TWiCS is partially supported by a grant from the Boeing Company. TWiCS members support each other and have fun through hands-on activities, discussions, outside speakers, tutoring sessions, field trips, and social events.

Each faculty member has a research program that often involves students, including some who sponsor summer projects or other for-credit or paid opportunities. Faculty from CS and beyond give opportunities for cross-disciplinary research including Mathematical Biology, Data Science, and Visual Design.

There are also opportunities to work with faculty and other students in the form of employment as tutors or graders for lower-level computer science courses.

DEPARTMENTAL HONORS IN COMPUTER SCIENCE

Honors in Computer Science may be earned by:

1. Maintaining an overall grade point average of 3.5,
2. Maintaining a major grade point average of 3.5,
3. Scoring at or above the 80th percentile on the senior exam,
4. Demonstrating excellence in scholarship by producing a scholarly paper or project, and
5. Receiving the approval of a majority of the regular faculty in computer science.

DEGREE REQUIREMENTS:

Dialogues Requirements: 42-61 Credits

Missouri Statute: 0-4 credits

Bachelor of Science Requirement: 7-8 Credits

Courses listed in the Bachelor of Science Requirement may double count with minors or second majors, but may not double count with Dialogues requirements nor be used elsewhere in the major.

- **MATH 263 - Analytic Geometry and Calculus II** Credit(s): 4
- Three or more credits from the following list:
 - Any STAT, MATH, or DATA coursework at the 300+ level, or
 - **DATA 222 - Data Science** Credit(s): 3
 - **MATH 200 - Foundations of Mathematics** Credit(s): 3
 - **MATH 264 - Analytic Geometry and Calculus III** Credit(s): 3
 - **STAT 250 - Statistical Computing** Credit(s): 3
 - Any AGSC, BIOL, CHEM, or PHYS course

MAJOR REQUIREMENTS:

The Computer Science major consists of three (3) parts: Required Support, Capstone Experience, and Major Requirements. Each student must complete all three parts.

Part I: Required Support: 12-16 Credits

Courses listed in Required Support may double count for Dialogues, minors, or other campus requirements, but may not double count with requirements listed elsewhere in this major. Students should work with their advisor to pick the right set of courses for their future goals.

- **MATH 198 - Analytic Geometry and Calculus I** Credit(s): 5

OR (MATH 187 - Calculus I with Precalculus Review I Credit(s): 4
and MATH 197 - Calculus I with Precalculus Review II Credit(s): 4)

- **STAT 190 - Basic Statistics** Credit(s): 3
OR **STAT 290 - Statistics** Credit(s): 3
- Any course that meets the Stem: Lab Science Perspective

Part II: Capstone Experience for Computer Science: 3 Credits

Each student is required to complete and present a Capstone experience that demonstrates the ability to work independently and to integrate and expand knowledge gained in major coursework. Prior approval and supervision by computer science faculty is required. While it is anticipated that most students present a substantial software development project, this could be interpreted quite broadly. Avenues could include an internship, research class, or a project directed by a faculty member in another discipline, employer, or an outside organization, such as an REU or work at ATSU.

Students with senior status who have completed CS 310 may request approval to allow at least two credits of a computationally-intensive High-Impact Experience, including CS 471/472, CS 473, CS 485, etc., to substitute for CS 497 in the CS major, provided that the project includes a written document, reflective component, and presentation.

Course credits may only be used once in the CS major. All Computer Science majors must complete CS 495.

Double-majors (especially in MATH, STAT, or DATA) may be able to build a project that fulfills the requirements of both major's capstone requirements, but should talk to a CS faculty member early to ensure an appropriate project.

- **CS 495 - Senior Computer Science Seminar** Credit(s): 1
- **CS 497 - Computer Science Capstone Experience** Credit(s): 2

Part III: Major Requirements: 42 Credits

Core Courses: 33 Credits

- **CS 100 - Computer Science Seminar** Credit(s): 1
- **CS 180 - Foundations of Computer Science I** Credit(s): 4
- **CS 181 - Foundations of Computer Science II** Credit(s): 3
- **CS 191 - Computing Structures** Credit(s): 3
- **CS 260 - Object-Oriented Programming and Design** Credit(s): 3
- **CS 291 - Models of Computation** Credit(s): 3
- **CS 310 - Data Structures and Algorithms** Credit(s): 3
- **CS 315 - Internet Programming** Credit(s): 3
- **CS 333 - Computer Systems Architecture: Design and Simulation** Credit(s): 4
- **CS 345 - Cyberethics** Credit(s): 3
- **CS 370 - Software Engineering** Credit(s): 3

Systems Elective: 3 Credits

Choose one course from the following list:

- **CS 390 - Operating Systems** Credit(s): 3
- **CS 420 - Compilers** Credit(s): 3
- **CS 435 - Parallel and Distributed Processing** Credit(s): 3
- **CS 455 - Computer Security Fundamentals** Credit(s): 3
- **CS 470 - Computer Networks** Credit(s): 3

General CS Electives: 6+ Credits

Choose courses from the following list to total six (6) or more credits:

- Additional Systems elective courses listed above
- **CS 325 - Introduction to Bioinformatics** Credit(s): 3
- **CS 380 - Programming Languages** Credit(s): 3
- **CS 430 - Database Systems** Credit(s): 3
- **CS 460 - Computer Graphics** Credit(s): 3
- **CS 480 - Artificial Intelligence** Credit(s): 3
- **CS 484 - Topics in Computer Science** Credit(s): 1 to 3
- **CS 490 - Automata Theory and Formal Languages** Credit(s): 3
- **DATA 520 - Seminar: Data Mining and Multivariate Statistics** Credit(s): 3

Electives to Total: 120 Credits

Suggested Courses:

Transfer students majoring in Computer Science must complete at least 18 credits in the major at Truman, including 15 credits at the 300 level or higher.

Internships:

An internship is a valuable part of the computer science learning experience for many students. As described above, some internships may fulfill the capstone project requirement in the major. Many students participate in additional internships at a variety of levels, sometimes after just one year at Truman.

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